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Phongsakon Mark Konrad

Curriculum Vitae

Summary

I want to contribute to genuine machine intelligence, systems that understand the world and work out its mechanisms the way science taught us to. I am not sure the dominant paradigm gets there, and scaling laws have never quite sat right with me. My work asks one question, whether the learning process that scaling relies on actually installs internal representations capable of genuine counterfactual reasoning, or only representations that imitate it. I study this through counterfactual reasoning and the geometry and identifiability of learned representations, and causal representation learning is home territory. I work phenomenon first and run the full research lifecycle myself, from problem formulation and experimental design to pipeline engineering, manuscript writing, and venue selection. I am a final year BSc Software Engineering student at SDU, admitted to the MPhil in Machine Learning and Machine Intelligence at the University of Cambridge for 2026, and I bring an unconventional background (four years of military service, three startups, a semester at HKUST) that shapes how I pick problems worth solving.

Education

2026– **MPhil, Machine Learning and Machine Intelligence**, *University of Cambridge*, Cambridge, UK, *admitted, begins Oct 2026*

2025 **Exchange Semester, Computer Science & Engineering**, *HKUST*, Hong Kong SAR
Specialisation courses in Machine Learning (COMP4211), Deep Learning in Computer Vision (COMP4471), Large Language Models (COMP4901B), and Reinforcement Learning (COMP4901Z), plus the postgraduate Data Visualisation (COMP6411D).

2023–2026 **BSc, Software Engineering**, *University of Southern Denmark*, Sønderborg, DK, GPA $\approx$ 10.6/12 ($\approx$ 3.7/4.0)
Expected June 2026. BSc thesis with T. L. Adam, *The Trader's Trinity: Forecasting Models, RL Agents, and LLM Judges for Day-Ahead Markets*.

Selected Publications

2026 **P. M. Konrad**, T. Tanyel, S. Ayvaz. *Self-Reports Do Not Identify Self-Models: An Identifiability Test for Counterfactual Reports*. Accepted (**oral**), *PhilML Workshop, ICML 2026*. [PDF]
In short. Self-reports are evidence about behaviour in a prompt, not proof of a self-model. Wrong-source demonstrations pull affect-state reports toward the source family unless mechanism binding holds.

2026 **P. M. Konrad**, T. Tanyel, S. Ayvaz. *Decoded but Unused: Instruction Tuning Routes Moral Framing into the Judgment Readout*. Accepted (poster), *ICML 2026 Workshop on Mechanistic Interpretability*. [PDF]
In short. Moral framing is linearly decodable in pretrained models but only causally routed to judgment after instruction tuning.

2026 **P. M. Konrad**, T. Tanyel, S. Ayvaz. *A Path Already Walked: On Inheriting Network-Neuroscience Tools for Mechanistic Interpretability*. Accepted (virtual poster), *ICML 2026 Workshop on Mechanistic Interpretability*. [PDF]
In short. Mech-interp should import the graph vocabulary of network neuroscience (modularity, rich clubs, motifs) under explicit contracts.

2026 N.-T. Thinh, Y. Du, **P. M. Konrad**, A. Narechania. *Fact-check Your Information (FYI): A Design Probe to Understand How People Actually Fact-check Data-Driven Articles*. Accepted, *IEEE VIS 2026*. [PDF]
In short. A browser-extension design probe with 22 readers finds three human and AI workflow archetypes for fact-checking data-driven journalism, with visualization as the main way people audit AI conclusions.

Research Experience

Jan 2026– **Research Collaborator**, *HKUST DataVISards group*, Hong Kong SAR
Machine learning and data-visualisation research.

Jan 2026– **Research Collaborator**, *SDU Data & Intelligence Lab*, Sønderborg, DK
Independently initiate and drive ML research from end to end, from problem formulation and experimental design to pipeline development, manuscript writing, and venue selection. Co-author with Assoc. Prof. Serkan Ayvaz across interpretability, NLP, medical imaging, and remote sensing.

Sep 2024–Dec 2025 **Research Assistant**, *SDU Data & Intelligence Lab*, Sønderborg, DK
Built ML pipelines end to end and contributed to study design, literature reviews, model development, and manuscript preparation. Supported grant proposals and multimodal dataset management.

Full Publication List

Peer-reviewed and accepted

Interpretability and safety (workshop)

1. **P. M. Konrad**, T. Tanyel, S. Ayvaz (2026). *Self-Reports Do Not Identify Self-Models: An Identifiability Test for Counterfactual Reports*. Accepted (**oral**), *PhilML Workshop, ICML 2026*. [PDF]
2. **P. M. Konrad**, T. Tanyel, S. Ayvaz (2026). *Decoded but Unused: Instruction Tuning Routes Moral Framing into the Judgment Readout*. Accepted (poster), *ICML 2026 Workshop on Mechanistic Interpretability*. [PDF]
3. **P. M. Konrad**, T. Tanyel, S. Ayvaz (2026). *A Path Already Walked: On Inheriting Network-Neuroscience Tools for Mechanistic Interpretability*. Accepted (virtual poster), *ICML 2026 Workshop on Mechanistic Interpretability*. [PDF]

AI for software engineering (workshop)

4. **P. M. Konrad**, T. L. Adam, R. Terrenzi, S. Ayvaz (2026). *Architecture Without Architects: How AI Coding Agents Shape Software Architecture*. Accepted, SAGAI Workshop, IEEE ICSA 2026. arXiv:2604.04990
5. T. L. Adam, **P. M. Konrad**, R. Terrenzi, F. G. Lukas, R. Yilmaz, K. Sierszecki, S. Ayvaz (2026). *CAKE: Cloud Architecture Knowledge Evaluation of Large Language Models*. Accepted, KDA-AI Workshop, IEEE ICSA 2026. arXiv:2604.05755
6. R. Terrenzi, **P. M. Konrad**, T. L. Adam, S. Ayvaz (2026). *A Reference Architecture for Agentic Hybrid Retrieval in Dataset Search*. Accepted, SAML Workshop, IEEE ICSA 2026. arXiv:2604.16394

Conference proceedings

7. N.-T. Thinh, Y. Du, **P. M. Konrad**, A. Narechania (2026). *Fact-check Your Information (FYI): A Design Probe to Understand How People Actually Fact-check Data-Driven Articles*. Accepted, *IEEE VIS 2026*. [PDF]
8. **P. M. Konrad**, T. Tanyel, S. Ayvaz (2025). *Beyond Major Floods: Deep Learning for Detecting Shallow Water Inundation in Agricultural Areas*. *KES 2025*, Procedia Computer Science, 270, 301–310. [open access]

Preprints and works in progress

Interpretability and safety

9. **P. M. Konrad**, T. Tanyel, S. Ayvaz (2026). *Routing Subspaces: Auditing Evaluation-to-Deployment Mismatch in Fine-Tuned Language Models*. Preprint. [PDF]
10. **P. M. Konrad**, T. Tanyel, S. Ayvaz (2026). *Acceptance Cards: A Four-Diagnostic Standard for Safe Fine-Tuning Defense Claims*. Preprint. arXiv:2605.10575
11. **P. M. Konrad**, T. L. Adam, A. C. H. Merrild, R. de Rosa, R. Terrenzi, T. Tanyel, S. Ayvaz (2026). *The Open-Box Fallacy: Why AI Deployment Needs a Calibrated Verification Regime*. Preprint. arXiv:2605.10601
12. **P. M. Konrad**, Y. Du, S. Ayvaz (2026). *Grounded Auditing as an Evaluation Policy: A Matched-Action Protocol and Stress Test*. Preprint. [PDF]
13. **P. M. Konrad**, S. Ayvaz (2026). *When Does Chain-of-Thought Improve Safety? Evidence from 18 Models across 5 Families*. Preprint. [PDF]
14. **P. M. Konrad**, Y. Du, T. Tanyel, S. Ayvaz (2026). *How Much of a Probe Direction Lives in the Input Embedding? An Audit of Four Published Claims*. Preprint. [PDF]
15. **P. M. Konrad**, Y. Du, T. Tanyel, S. Ayvaz (2026). *Abstention and Refusal Are Separable Directions in the Residual Stream*. Preprint. [PDF]

Applied ML and other

15. **P. M. Konrad**, A.-A. Popa, Y. Sabzehmeidani, L. Zhong, M. Tripathy, A. Constantinescu, E. A. Liehn, S. Ayvaz (2025). *Challenges in Deep Learning-Based Small Organ Segmentation: A Benchmarking Perspective for Medical Research with Limited Datasets*. Under revision, *Biomedical Signal Processing and Control*. arXiv:2509.05892
16. **P. M. Konrad**, C. H. Kunstmann-Olsen, J. Fiutowski, S. Ayvaz (2025). *Non-Destructive Prediction of Fruit Ripeness*

and Firmness Using Hyperspectral Imaging and Lightweight Machine Learning Models. Preprint. arXiv:2604.22788

Teaching

2026 **Teaching Assistant, Artificial Intelligence (BSc)**, *University of Southern Denmark*
Lead exercise sessions and student supervision and design hands on lab assignments. Built an interactive AI course with step by step animations, used as supplementary material.

Academic Service

2025– **Student Member**, Educational Committee, Software Engineering Programme, SDU. Represent students in curriculum and quality assurance discussions on course content, assessment, and study environment.

Honors and Awards

2026 **Nominated, Best Thesis Award**, Faculty of Engineering, University of Southern Denmark.

2026 **William Demant Fonden Grant** supporting graduate study.

2024, 2025 **Top 10**, Danish National Championship in AI (DM i AI), competing solo against 50+ teams.

2026 **Nominated, Best Startup Award** for DreamBear (SaturoLabs).

2023 **1st place**, SDU Case Competition (48 hour sustainability challenge, Danfoss and Linak cases).

2021 **Commendation and performance bonus**, Bundeswehr, for exemplary and independent service.

Professional Experience

2026– **Founder**, *SaturoLabs*, Denmark
Products at the intersection of AI and software engineering. Live products include claudeboyz.com, getproofz.com, and dreambear.app.

2024 **CTO & Co-Founder**, *Tutora ApS*, Sønderborg, DK
Led development of the company's web application from end to end.

2022–2024 **Co-CEO & Co-Founder**, *Yeager GmbH*, Remote
Built *stabil.ai*, an AI powerlifting training app with personalised plans via MRV and MEV and real time feedback.

2017–2021 **Staff Duty Soldier**, *Bundeswehr (German Armed Forces)*, Glücksburg, DE
Led a small HR team administering 600+ soldiers. Twice decorated.

Technical Skills

Languages Python, JavaScript, SQL

ML PyTorch, Hugging Face, TransformerLens, scikit-learn, pandas, NumPy, Optuna, Weights & Biases

Tooling Git, Docker, Google Cloud Platform, React, React Native, Node.js

Languages

Fluent English (professional), German (native), Thai (native)

Basic Danish